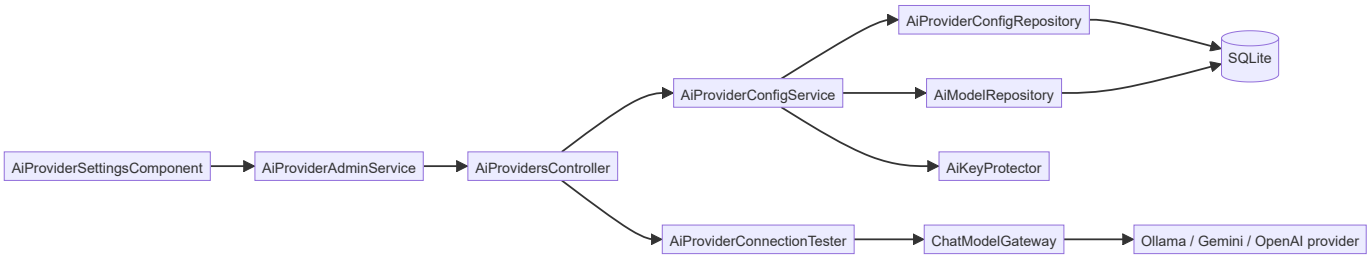
TFA Settings – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Settings · **Child Module:** 08.4 TFA Settings **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

TFA Settings is `AiProvidersController` over `AiProviderConfigService`, `AiProviderConnectionTester`, `AiProviderConfigRepository` and `AiModelRepository`. It stores one `AiProviderConfigs` row per AI provider (Ollama, Gemini, OpenAI) and the list of selectable `AiModels`. API keys are encrypted with ASP.NET Core Data Protection (`AiKeyProtector`) and never returned. The connection test goes through the Assistant's `IChatModelGateway` (`ChatModelGateway`) to the provider clients in `Integrations/Ai/Providers/`.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Settings/`	`AiProvidersController.cs`
Service	`TaskFlow.Api/Services/Settings/TfaSettings/ProviderConfig/`	`IAiProviderConfigService.cs`, `AiProviderConfigService.cs`
Service (test)	`TaskFlow.Api/Services/Settings/TfaSettings/ConnectionTest/`	`IAiProviderConnectionTester.cs`, `AiProviderConnectionTester.cs`
Repository	`TaskFlow.Api/Repositories/Settings/TfaSettings/ProviderConfig/`	`IAiProviderConfigRepository.cs`, `AiProviderConfigRepository.cs`

Layer	Path	Files
Repository	`TaskFlow.Api/Repositories/Settings/TfaSettings/AiModels/`	`IAiModelRepository.cs`, `AiModelRepository.cs`
DTOs	`TaskFlow.Api/Dtos/Settings/TfaSettings/`	`AiProviderDtos.cs` (`AiProviderOptionDto`, `AiProviderConfigDto`, `UpdateAiProviderRequest`, `UpdateAiProvidersRequest`, `AiProviderTestResultDto`)
Validators	`TaskFlow.Api/Dtos/Settings/TfaSettings/Validators/`	`UpdateAiProvidersRequestValidator.cs` (also `UpdateAiProviderRequestValidator`)
Entities	`TaskFlow.Api/Models/Entities/`	`AiProviderConfig.cs`, `AiModel.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`AiProviderKind.cs`, `ProviderFailureKind.cs`
Constants	`TaskFlow.Api/Common/Constants/`	`AiProviderCatalog.cs`
AI integration	`TaskFlow.Api/Integrations/Ai/`	`KeyProtection/AiKeyProtector.cs`, `KeyProtection/KeyRingProtectionCheck.cs`, `Registry/ChatModelProviderRegistry.cs`, `Providers/OllamaProvider.cs`, `Providers/GeminiProvider.cs`, `Providers/OpenAiProvider.cs`, `Providers/ProviderHttp.cs`, `Providers/ProviderFailure.cs`, `Providers/AiProviderRuntime.cs`, `Providers/ChatModelTypes.cs`
Gateway (Assistant module)	`TaskFlow.Api/Services/Assistant/ModelGateway/`, `Services/Assistant/Chat/`	`ChatModelGateway.cs`, `ProviderFailureMessages.cs`
Seeders	`TaskFlow.Api/Persistence/Seeders/`	`AiProviderSeeder.cs`
Config	`TaskFlow.Api/Config/`	`AiProviderOptions.cs`, `IntegrationRegistration.cs`
Migrations	`TaskFlow.Api/Persistence/Migrations/`	`20261018183210_AddAssistant.cs` (`AiProviderConfigs`), `20261019163241_AddAiModels.cs` (`AiModels`)

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/settings/tfa-settings/pages/tfa-settings.component.ts`	`TfaSettingsComponent` wrapper

Type	Path	Purpose
Child component	`src/app/modules/settings/tfa-settings/components/ai-provider-settings.component.{ts,html}`, `ai-provider-settings.model.ts`	`AiProviderSettingsComponent`, `ProviderForm`, `ADD_MODEL`, `CLOUD_SERVERS`
API service	`src/app/modules/settings/tfa-settings/services/ai-provider-admin.service.ts`	`AiProviderAdminService.getConfig()/save()/removeModel()/test()`
Route / Guard	`src/app/modules/settings/settings.routes.ts` (`settings/tfa`), `src/app/shared/auth/auth.guard.ts`	`permissionGuard` with `SettingsAiView`
Menu	`src/app/layouts/nav-items.ts`	"TFA Settings" item when `SettingsAiView` is held

5. API & Logic

5.1 API Endpoints

Controller `AiProvidersController`, `[Route("api/v1/assistant/providers")]`, class `[Authorize]`.

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/assistant/providers/config`	Every provider with status. Key never returned	`settings.ai.view`	–	`IReadOnlyList<AiProviderConfigDto>`
2	PUT	`/api/v1/assistant/providers/config`	Save provider settings	`settings.ai.edit`	`UpdateAiProvidersRequest`	`IReadOnlyList<AiProviderConfigDto>`
3	DELETE	`/api/v1/assistant/providers/{kind}/models?name=`	Remove an administrator-added model	`settings.ai.model.remove`	Query `name`	`IReadOnlyList<AiProviderConfigDto>`
4	POST	`/api/v1/assistant/providers/{kind}/test`	One tiny request to the provider	`settings.ai.test`	–	`AiProviderTestResultDto`

`{kind}` binds to `AiProviderKind` (`Ollama`, `Gemini`, `OpenAI`). Related (Assistant module): `GET /api/v1/assistant/providers` (`assistant.chat`) returns `AiProviderOptionDto` for configured providers only, through `GetAvailableAsync`.

5.2 Request / Response Samples

```
// PUT /api/v1/assistant/providers/config
{ "providers": [
  { "kind": "Ollama", "isEnabled": false, "modelName": "llama3.2:3b", "baseUrl":
"http://localhost:11434" },
  { "kind": "Gemini", "isEnabled": true, "modelName": "gemini-2.0-flash", "apiKey": "<secret>"
},
  { "kind": "OpenAI", "isEnabled": false, "modelName": "gpt-4o-mini", "clearKey": true }
] }
```

```
// 200 OK (one element)
{ "kind": "Gemini", "displayName": "Google Gemini", "locality": "Cloud", "requiresKey": true,
"isEnabled": true, "modelName": "gemini-2.0-flash", "baseUrl": null, "keySet": true,
"keyLast4": "x9Qa", "isConfigured": true, "keyReadable": true, "models": ["gemini-2.0-flash",
"gemini-2.5-flash", "gemini-2.5-pro"], "customModels": [] }
```

```
// POST /api/v1/assistant/providers/Ollama/test -> 200 OK
{ "ok": false, "message": "Ollama is not running or cannot be reached. Ask an administrator to
start it, or choose another model.", "failureKind": "Unreachable" }
```

5.3 Service Layer Logic

Method: `AiProviderConfigService.GetAdminConfigAsync()`

- For each `AiProviderCatalog.All` kind, `ToDto` builds the response:
  - `Models` = all `AiModels` of that kind, plus the chosen model if it is missing
  - `CustomModels` = the non-built-in ones
  - `BaseUrl` only for Ollama (default `http://localhost:11434`)
  - `KeySet` and `KeyLast4`
  - `KeyReadable` = the stored key can still be `Unprotect`-ed
  - `IsConfigured` = enabled, has a model, and has either a URL (Ollama) or a decryptable key

Method: `AiProviderConfigService.SaveAsync(int userId, UpdateAiProvidersRequest)`

1. Loads all tracked config rows. For each update:
   - a null `Kind`, `IsEnabled` or `ModelName` gives `RequestValidationException("providers", "Every provider setting must be provided.")`
   - a trimmed model name that is empty or over 100 characters gives "The {Provider} model name must be 1 to 100 characters."
2. Creates the row if it is missing. Sets `IsEnabled` and `ModelName`. When the provider is enabled, `RememberModelAsync` adds a non-built-in `AiModel` if the name is new.
3. `BaseUrl`: for Ollama, the normalised value (trimmed, trailing `/` removed) or the default. For cloud providers, `null`.
4. A non-empty `ApiKey` on Ollama gives "Ollama does not use an API key.". Otherwise `EncryptedApiKey = _keyProtector.Protect(key)` and `KeyLast4` = the last 4 characters. `ClearKey == true` nulls both.
5. Sets `UpdatedByUserId` and `UpdatedAt`. `SaveKeepingAnyDuplicateModelAsync` handles a duplicate model race: on `UniqueConstraintException` it detaches the duplicate added models and saves again.
6. The frontend sends `isEnabled = true` only for the selected provider, so one server is active. The backend itself allows several enabled providers.

Method: `AiProviderConfigService.RemoveModelAsync(kind, name)`

- Not found gives `NotFoundException("That model is not in the list.")`. `IsBuiltIn` gives a 400 "A built-in model cannot be removed.". `UsesModelAsync` gives a 400 "That model is the one saved for its server. Choose another model and save before removing it.". Otherwise it removes the model and returns the config.

Method: `AiProviderConfigService.GetRuntimeAsync(kind)`

- Returns `AiProviderRuntime(kind, model, baseUrl, decryptedKey)` or `null` when the provider is not configured. It logs a warning when an enabled provider's key cannot be decrypted.

Method: `AiProviderConnectionTester.TestAsync(kind)`

1. Builds a `ChatModelRequest` with the system text "This is a connection test. Reply with the single word OK." and the user message "Reply with OK.".
2. Calls `IChatModelGateway.CompleteAsync(kind, request)`. The gateway gets the runtime, throws `ProviderNotConfiguredException` when it is null, and calls the provider through `IChatModelProviderRegistry`.
3. On success, `Ok = true` with "The connection works.". `ProviderNotConfiguredException` gives `ProviderFailureMessages.NotConfigured(kind)`. `ProviderFailureException` gives `ProviderFailureMessages.For(kind, failure.Kind, failure.RetryAfter)` and its `FailureKind`. The test always returns `200`, never an error status.

Provider clients:

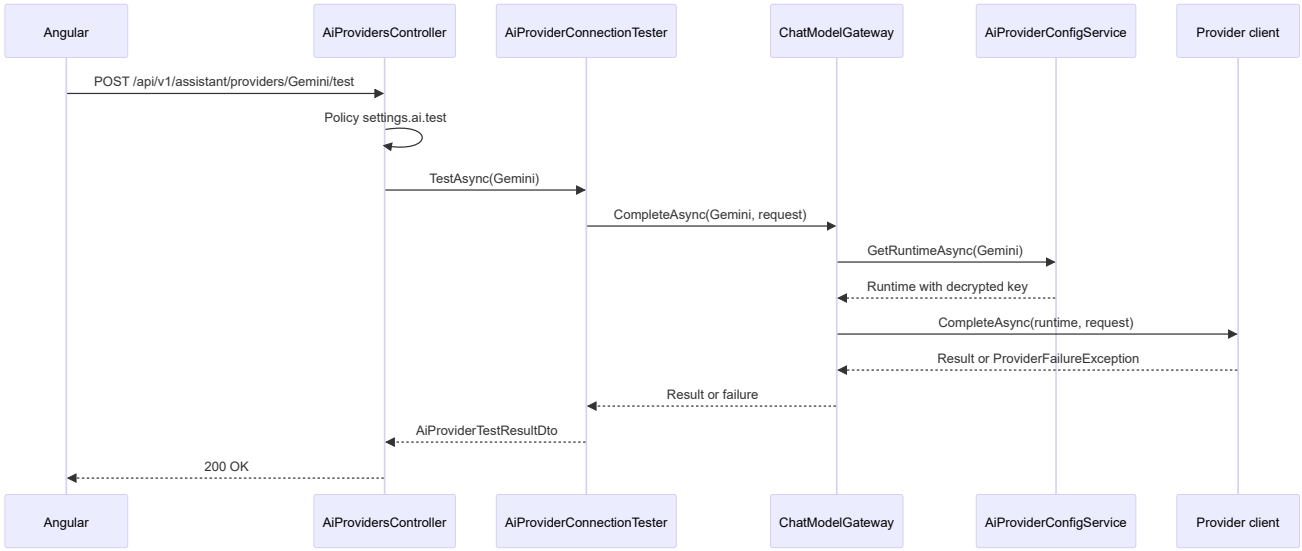
- `OllamaProvider` calls `{BaseUrl}/api/chat` with no key.
- `GeminiProvider` calls `v1beta/models/{model}:generateContent`, with the key in the `x-goog-api-key` header.
- `OpenAiProvider` calls `https://api.openai.com/v1/chat/completions`.

`ProviderHttp` maps a timeout to `Timeout` and an `HttpRequestException` to `Unreachable`. Status codes and provider error codes map to `RateLimited` (429), `InvalidKey` (401/403) and `ModelUnavailable` (404). Redirects are not followed.

5.4 Repository Layer

Method	Description	Query Notes
`AiProviderConfigRepository.GetAllAsync()`	All provider rows	`AsNoTracking`
`AiProviderConfigRepository.GetAllForUpdateAsync()`	Tracked rows for save	Tracked `List`
`AiProviderConfigRepository.GetAsync(kind)`	One row	`AsNoTracking`
`AiProviderConfigRepository.UsesModelAsync(kind, name)`	Is this the saved model	`AnyAsync`
`AiModelRepository.GetAllAsync()`	All models	`AsNoTracking`, built-in first, then by `Id`
`AiModelRepository.IsKnownAsync(kind, name)`	Stored or staged in this request	Checks `Set.Local` then the database
`AiModelRepository.GetForUpdateAsync(kind, name)`	Tracked model to remove	`FirstOrDefaultAsync`
`AiModelRepository.DropAddedModelsStoredElsewhereAsync()`	Race recovery	Detaches `Added` entries that now exist

5.5 Sequence Diagram



6. Database Design

Table	Column	Type	Constraints	Description
`AiProviderConfigs`	`Id`	int	PK	Row id
`AiProviderConfigs`	`Kind`	string(16)	Unique (enum as string)	`Ollama`, `Gemini`, `OpenAI`
`AiProviderConfigs`	`IsEnabled`	bool	Not Null	Provider switched on
`AiProviderConfigs`	`ModelName`	string(100)	Not Null	Chosen model
`AiProviderConfigs`	`BaseUrl`	string(300)	Nullable	Ollama only
`AiProviderConfigs`	`EncryptedApiKey`	string	Nullable	Data Protection ciphertext, never returned
`AiProviderConfigs`	`KeyLast4`	string(4)	Nullable	Display hint
`AiProviderConfigs`	`UpdatedByUserId`	int?	– (no FK)	Last editor
`AiProviderConfigs`	`UpdatedAt`	DateTime	Not Null	UTC
`AiModels`	`Id`	int	PK	Row id
`AiModels`	`Kind`	string(16)	Unique with `Name`	Provider
`AiModels`	`Name`	string(100)	Not Null, Unique with `Kind`	Model name
`AiModels`	`IsBuiltIn`	bool	Not Null	From `AiProviderCatalog.BuiltInModels`

Relationships: no FK. `AiModels.Kind` and `AiProviderConfigs.Kind` are matched by value.

Seed data: `AiProviderSeeder` inserts the built-in models and one config row per provider. Ollama is enabled only on a fresh install (no users yet).

7. Security & Authorization

- **Authorization:** class `[Authorize]` plus one `[RequirePermission]` per action (`settings.ai.view`, `settings.ai.edit`, `settings.ai.model.remove`, `settings.ai.test`).
- **Secrets:**
  - API keys are encrypted with Data Protection (purpose `TaskFlow.AiProviderKeys`). The key ring lives at `Storage:Root` / `AiProviders:KeyRingPath` and is DPAPI-protected on Windows.
  - At startup, `KeyRingProtectionCheck` warns about unprotected master keys.
  - Keys are never returned. Provider logs contain only provider, failure kind and HTTP status.
  - The Ollama URL validator blocks link-local and cloud-metadata hosts (SSRF guard).
- **Frontend:** `permissionGuard` on `settings/tfa`, the menu item in `buildNavigation()`, and in-component checks `canEdit`, `canTest` and `mayRemoveModel`.

Permission Key	Endpoint(s)	UI Element Controlled
`settings.ai.view`	`GET /api/v1/assistant/providers/config`	Menu "TFA Settings", route `/settings/tfa`, section visibility
`settings.ai.edit`	`PUT /api/v1/assistant/providers/config`	"Save TFA settings" button
`settings.ai.test`	`POST /api/v1/assistant/providers/{kind}/test`	"Test connection" button
`settings.ai.model.remove`	`DELETE /api/v1/assistant/providers/{kind}/models`	"Remove this model from the list" button
`settings.ai.manage` (legacy)	–	Not shown. Expands to the four `settings.ai.*` keys for roles that still hold it

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Providers null / duplicate kind	`UpdateAiProvidersRequestValidator`	400	Providers are required. / Each provider can only appear once.
Model name blank	`UpdateAiProviderRequestValidator`	400	Model name cannot be blank.
Bad / metadata URL, > 300	Validator	400	The server address must be a valid http or https URL, and not a link-local or cloud metadata address.
Key with whitespace, > 500	Validator	400	The API key cannot contain spaces.

Rule	Layer	HTTP Code	Error Message
New key and clearKey together	Validator	400	Provide a new key or clear the key, not both.
Missing provider field	`AiProviderConfigService`	400	Every provider setting must be provided.
Model name length	`AiProviderConfigService`	400	The {Provider} model name must be 1 to 100 characters.
Key sent for Ollama	`AiProviderConfigService`	400	Ollama does not use an API key.
Remove: name missing	`AiProvidersController`	400	The model name is required.
Remove: not in list	`AiProviderConfigService`	404	That model is not in the list.
Remove: built-in	`AiProviderConfigService`	400	A built-in model cannot be removed.
Remove: saved model	`AiProviderConfigService`	400	That model is the one saved for its server. Choose another model and save before removing it.

The connection test never returns an error status; failures come back in `AiProviderTestResultDto` (`ok: false`, `message`, `failureKind`). Shared rules are in the parent document.

9. Integrations / Side Effects

- **AI Assistant:** `AssistantService` and `AssistantController` use `GetAvailableAsync` for the model selector and `IChatModelGateway` for calls. Assistant knowledge page: `Services/Assistant/Knowledge/tfa-settings.md`.
- **External providers:** the test and all TFA calls go to Ollama (`{BaseUrl}/api/chat`), Gemini or OpenAI as described in 5.3.
- **Activity Log:** none (only `UpdatedByUserId` / `UpdatedAt` on each provider row).

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`AiProviders:KeyRingPath`	Data Protection key-ring folder, relative to `Storage:Root`	`dp-keys`
`AiProviders:TimeoutSeconds`	Timeout of one provider HTTP call	`60`
`Storage:Root`	Root folder for the key ring (shared, see parent)	`""`

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Settings/TfaSettings/ProviderConfig/AiProviderConfigServiceTests.cs`	Configured state per provider; model lists and adding, removing and racing custom models; only configured providers are listed for users; keys are encrypted, shown as the last four characters, kept when empty, replaced or cleared; an unreadable key is reported and logged without key material; validator rules
Unit	`tests/TaskFlow.UnitTests/Settings/TfaSettings/ProviderConfig/AiProviderSeederTests.cs`	A fresh database gets every built-in model and Ollama on with cloud providers off; an upgraded install gets every provider off; re-running adds nothing and never overwrites saved values
Unit	`tests/TaskFlow.UnitTests/Integrations/Ai/KeyProtection/AiKeyProtectorTests.cs`, `KeyRingProtectionCheckTests.cs`	Keys round-trip and are not stored as plain text; another purpose cannot read them; a replaced key ring gives null; the key-ring check counts unprotected keys and handles missing, empty and unreadable folders and files
Unit	`tests/TaskFlow.UnitTests/Integrations/Ai/Providers/ChatProviderTests.cs`	Ollama, Gemini and OpenAI clients: text and tool-call round trips, key headers, failure classification (unreachable, model unavailable, rate limited, timeout), no key or request text in failure messages, and the provider registry
Unit	`tests/TaskFlow.UnitTests/Assistant/ModelGateway/ProviderFailureHandlingTests.cs`	Failure messages suggest another model, use the retry time and never include raw provider text; a failure never falls over to another provider; the per-user limit of 20 requests a minute
Integration	`tests/TaskFlow.IntegrationTests/Settings/TfaSettings/AssistantProviderEndpointsTests.cs`	Non-administrators get 403 and no token gets 401; a saved key never appears in a response or log; on Windows the key ring is protected with DPAPI; invalid input gets 400; test connection reports success and classified failures; only an added, unused model can be removed
Integration	`tests/TaskFlow.IntegrationTests/Settings/TfaSettings/AiDefaultsOnFirstRunTests.cs`	A new database has the model list and Ollama ready, so every signed-in user can use TFA on first run
Integration	`tests/TaskFlow.IntegrationTests/Integrations/Ai/ProviderRedirectTests.cs`	A provider response that redirects is not followed
Integration	`tests/TaskFlow.IntegrationTests/Common/KeyRingStartupWarningTests.cs`	At startup a plain key in the key ring logs a warning with the fix and no key material; empty or protected rings give no warning; an unreadable key file does not stop startup
Integration (support)	`tests/TaskFlow.IntegrationTests/TestSupport/ScriptedChatProvider.cs`	Not a test. In-process fake provider for each `AiProviderKind`, registered by `TaskFlowApiFactory`; records the calls it receives
Front end unit	`src/app/modules/settings/tfa-settings/components/ai-provider-settings.component.spec.ts`	Hidden from non-administrators; server dropdown and per-server fields; model lists, adding and removing models; which server TFA uses; key handling (never shown, kept, replaced, removed, unreadable); input validation and save errors; test connection saves first and shows each failure

Test Type	File / Location	Covers
Front end unit	`src/app/modules/settings/tfa-settings/pages/tfa-settings.component.spec.ts`	The "TFA settings" screen holds the model settings; without `settings.ai.manage` nothing is shown or requested

> The DPAPI test in `AssistantProviderEndpointsTests.cs`, all of `KeyRingStartupWarningTests.cs` and four tests in `KeyRingProtectionCheckTests.cs` use `[WindowsOnlyFact]` and run only on Windows.

12. Known Limitations / Gaps found

- The backend allows several providers to be enabled at once. Only the frontend enforces "one active server".
- **TFA view-only inputs:** without `settings.ai.edit`, the Model, Server, address and API-key inputs and the Remove/Keep key buttons stay editable. Only Save is hidden (audit).
- **TFA test with unsaved changes:** `test()` calls `save()` first when `needsSave()` is true. A user with `settings.ai.test` but not `settings.ai.edit` gets a 403 from the save, and the test does not run.

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